Relational Actualism: Irreversible Events as the Primitive Basis of Physical Reality

Relational Actualism — Paper V: The Complete Framework

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April 2026

Abstract

Relational Actualism (RA) is a framework in which irreversible on-shell actualization events — physical interactions producing a permanent increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > \Delta S^* = 0.60069\,\text{nats}$ — are the primitive physical reality. Spacetime is the growing directed acyclic graph (DAG) of these events; time is the process of graph growth; the arrow of time is encoded in the irreversibility of actualization itself.

From a single ontological commitment, without free parameters, RA derives: $\alpha_{\text{EM}}^{-1} = 137$ [LV]; $\alpha_s(m_Z) = 1/\sqrt{72}$ [CV]; Koide $K = 2/3$ [LV]; complete Standard Model spectrum [LV]; $\Lambda = 0$ [DR]; BMV null prediction [DR]; Hubble tension $H_0^{\text{Eridanus}} \approx 75.9\,\text{km/s/Mpc}$ [CV]; WIMP prohibition [DR]; Boltzmann Brain prohibition [DR]; minimum viable universe seed $N_{\min} = 2$ [DR]; universal decoherence timescale $\tau_d = 20\,\text{fs}$ at $300\,\text{K}$ [DR]; Kinematic Coherence Bound for quantum computing [AR]; Causal Firewall origin-of-life threshold (same $N_{\min} = 2$ as Big Bang) [DR].

Approximately 110 results are Lean 4 verified with zero sorry tags in the core files. All 76 KB claims are documented with explicit epistemic status. This paper is the hub of the series: Papers I–IV are self-contained derivations; Paper V provides the complete epistemic map, the coherence structure analysis, all predictions, all open problems, and the philosophical synthesis.

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1 The Founding Commitment

1.1 The Ontological Claim

The Relational Actualism Commitment:

An interaction constitutes a *physical event* if and only if it produces an irreversible increase in quantum relative entropy:

$$\Delta S(\rho \| \sigma_0) > \Delta S^* = 0.60069 \,\text{nats} \tag{1}$$

Such events write permanent directed edges into the growing causal DAG. The DAG is reality. Everything else is derived from it.

This is not a hypothesis about the nature of spacetime alongside other hypotheses. It is a proposal that irreversibility is the most fundamental fact about physical reality. The universe is not a stage on which events play out; the universe *is* the events.

1.2 Time as the Growing Graph

In standard physics, time is a parameter. Events are *located in* time. The laws of physics are equations that describe how quantities evolve *through* time. Time is the background.

In RA, this is inverted. Time *is* the growing of the causal graph. The past is the part of the graph that has been written. The future is what has not yet been written. The present is the process of the next edge being added.

Proper time along a worldline — the time measured by a clock — is the integer count of actualization events along that path. Relativistic time dilation is the statement that different worldlines accumulate events at different rates. The Lorentz factor $\gamma = dt/d\tau$ is the ratio of coordinate events to proper events.

The arrow of time is not explained by RA — it is what RA *is*. Actualization is by definition irreversible. The causal graph can only grow, never shrink. There is no equation of motion in RA that runs both forward and backward; the BDG criterion $S(v, C) > 0$ is not symmetric under time reversal.

1.3 The BDG Action

The dynamics are governed by the Benincasa–Dowker–Glaser action [5]:

$$S(v, C) = 1 - N_1 + 9N_2 - 16N_3 + 8N_4 \tag{2}$$

The integers $(1, -1, 9, -16, 8)$ are uniquely determined by the requirement that this operator converges to the d'Alembertian (Paper I, Section 2). They are not free parameters.

2 The Epistemic Map

2.1 Why Explicit Status Matters

The epistemic map is the most important methodological innovation in this series. Theoretical physics has a history of overconfident claims: frameworks that were presented as derived when key steps were argued, or as established when they were conjectured. RA attempts to reverse this tendency by documenting the epistemic status of every claim.

Status does not measure importance. Some of the most important results are [AR] or [DR]; some [LV] results are supporting lemmas. Status measures epistemic certainty.

2.2 Complete Lean 4 Verification Table

Table 1: Epistemic status categories.

Badge	Name	Definition
[LV]	Lean-Verified	Machine-checked in Lean 4. Zero sorry tags in core files. Certainty as high as formal mathematics allows.
[CV]	Computation-Verified	Certified numerical computation with explicit error bounds. Reproducible by running the public scripts.
[DR]	Derived	All steps explicit in the RA papers. Argument is complete; the result follows from the premises. Not machine-checked.
[AR]	Argued	Key steps identified and physically motivated. Some steps not fully explicit. Upgradeable with further work.
[OP]	Open	Precisely scoped gap with a stated path to resolution.

Table 2: All Lean 4 verified results. Files: RA_D1_Proofs.lean (main), RA_AmpLocality.lean (amplitude locality), RA_Spin2_Macro.lean (spin-2), RA_AQFT_Proofs_v10.lean (AQFT). One intentional sorry in a library adapter unrelated to any RA claim. Total: ≈ 110 verified results.

ID	Theorem	Statement
L01	LLC	$\Sigma_{\text{out}}(v) = \Sigma_{\text{in}}(v)$ at every DAG vertex
L02	Graph Cut	LLC holds on both sides of any causal severance
L03	Markov Blanket	Interior subgraph shielded from exterior
L04	Frame Independence	$S(U\rho U^\dagger \| \sigma_0) = S(\rho \| \sigma_0)$ for all Poincaré U
L05	Rindler Stationarity	$\frac{d}{dt} S(\alpha_t(\rho_\beta) \| \sigma_0) = 0$ under modular flow
L06	Rindler Thermal	ρ_β is valid density matrix (Hermitian, positive, tr $=$ 1)
L07	Causal Inv. (cond.)	$\mu_\sigma(S) = \mu_{\sigma'}(S)$ given amplitude locality
L08	α_{EM}	$144 - 7 = 137$ (`norm_num`)
L09	Koide	$K = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$
L10	Confinement depths	$L_{\text{gluon}} = 3$, $L_{\text{quark}} = 4$
L11	BDG Closure	5 topology types, 124 extensions verified
L12	Qubit Fragility	Min BDG score for electrons/photons $= 1$
O01	Amplitude Locality	$a(v\|C)$ depends only on $C \cap \text{past}(v)$
O02	Causal Invariance	$\mu_\sigma(S)$ invariant for any permutation
O10s	Spin-2 DOF	$5 - 1 - 1 - 1 = 2$ physical degrees of freedom for BDG metric excitation

2.3 Computation-Verified Results

Table 3: All computation-verified results. Scripts in the GitHub repository are public and reproducible.

ID	Result	Value	Reference
C01	$\alpha_s(m_Z) = 1/\sqrt{72}$	0.11785 (PDG: 0.1180)	d3_alpha_s_proof.py
C02	$P_{\text{acc}}(1)$	0.548436	BDG Monte Carlo
C03	ΔS^*	0.60069 nats4(irrational)	exact enumeration
C04	Spin-bath	$t^* = 0.274/g$	spin-bath model
C05	Hubble tension	$H_0^{\text{Eridanus}} = 75.9\,\text{km/s/Mpc}$	void geometry
C06	Proton mass	$m_p = \sqrt{8}\Lambda_{\text{QCD}}$ (0.08%)	BDG Regge

3 The Coherence Hub

3.1 Three Hub Results from April 2026

In April 2026, three previously open problems were closed in a single session. An audit of the new results revealed that each was a hub: each closed result enabled the closure or substantial progress of multiple other open problems. This section documents the hub structure.

Hub 1: D4U02-analytic. The analytic proof that $P(S = 1) > P(S = 0)$ via conditioning and modular arithmetic:

- Proved D4U02 analytically (OP→DR) after six years.
- Established dimensional uniqueness ($d = 4$ from modular barrier, new DR result).
- Opened new perturbative path for O03 (Bianchi on curved background).
- Motivated the $l_{\mathrm{RA}} = \ell_P$ identification from the dominant $k = 2$ term (O06).
- Provided a concrete question for Dowker on Lorentz emergence (O11).

Hub 2: Universal timescale $\tau_{\mathrm{act}} = \Delta S^*/\Gamma_{\text{on-shell}}$. The formula and its three-scale application:

- Closed O07 (BMV timescale) as a derived result.
- Closed O05 (Unruh detector) via L05+timescale.
- Closed O08 (biological τ_d) via timescale at $T = 300\,\mathrm{K}$.
- Upgraded P01 (BMV null) from AR to DR.
- Connected cosmological and biological timescales through one formula.

Hub 3: COSMO-SEED bootstrap. The $N_{\min} = 2$ condition:

- Established exact BDG arithmetic for Big Bang seeds.
- Identified $N_{\min} = 2$ = Causal Firewall origin-of-life threshold (same condition, two scales).
- Provided the exact Cosmological Natural Selection mechanism.
- Derived the BH mass hierarchy for viable universe production.

3.2 The Universal Timescale as the Central Hub

The formula $\tau_{\text{act}} = \Delta S^*/(\Gamma_{\text{on-shell}} \cdot \Delta S_{\text{per boson}})$ in five physical contexts:

$$T = 1\,\text{mK}: \quad \tau_{\text{act}} \approx 1.9 \times 10^4\,\text{s} \gg \tau_{\text{exp}} \quad \text{(BMV null)} \tag{3}$$

$$T = 300\,\text{K}: \quad \tau_d \approx 20\,\text{fs} = \tau_{\text{vib}} \quad \text{(classical chemistry)} \tag{4}$$

$$\text{Rindler}: \quad \tau_{\text{act}} \to \infty \quad (\text{KMS: } \Delta S_{\text{per photon}} = 0) \tag{5}$$

$$N = 2 \text{ seed}: \quad \tau_d = 0 \quad \text{(instantaneous bootstrap)} \tag{6}$$

$$\text{Spin-bath}: \quad t^* = 0.274/g \quad \text{(parameter-free)} \tag{7}$$

Fourteen orders of magnitude. No free parameters. One formula.

3.3 The $N_{\text{min}} = 2$ Identity in Full

The minimum viable universe seed (Paper III) and the Causal Firewall origin-of-life threshold (Paper IV) are the same condition: $N \geq 2$ causally connected vertices with a directed edge. BDG viability $V = 1 - N_1 + 9N_2 = 1 - 1 + 9 = +9$ in both cases.

These are not coincidences. They are the same geometric fact — the minimum non-Markovian causal structure in the Poisson-CSG model — applied at two different scales. The Big Bang and the first self-replicating molecule share the same fundamental causal requirement.

4 General Relativity and Cosmology

4.1 From LLC to Einstein Equations

The route to $G_{\mu\nu} = 8\pi G P_{\text{act}}[T_{\mu\nu}]$ with $\Lambda = 0$ uses four Lean-verified inputs:

1. L01 (LLC) gives conservation at every vertex.
2. In the flat/weak-field limit (D04, [DR]), LLC $\Rightarrow \nabla_\mu G^{\mu\nu} \equiv 0$.
3. L11 ($d = 4$ from SM closure) activates Lovelock's uniqueness theorem.
4. D03 (vacuum suppression) sets $\Lambda = 0$.

Combined with O10-spin2 ([LV]: 2 physical DOF for the BDG metric excitation) and amplitude locality (O01, [LV]), the Weinberg route [7] gives the unique coupling: $G_{\mu\nu} = 8\pi G P_{\text{act}}[T_{\mu\nu}]$.

The cosmological constant $\Lambda = 0$ because $P_{\text{act}}[\rho_{\text{vac}}] = 0$: the vacuum consists of off-shell virtual processes with $\Delta S = 0$, all below the ΔS^* threshold.

4.2 Black Hole Thermodynamics

$S_{\text{BH}} = A/(4\ell_P^2)$ follows from D10 ([AR], Bridge Lemma): L02 (LLC across severance) + L05-L06 (KMS state) $\Rightarrow$ the Jacobson thermodynamic derivation [6]. Status is AR because the l_{RA} identification (O06) is not yet a derived result; when O06 closes, D10 upgrades to DR automatically.

5 Quantum Information: The Kinematic Coherence Bound

Theorem L12 (Qubit Fragility) [LV]**:** Electrons and photons have minimum BDG stability score 1. (`structural_fragility`, RA_D1_Proofs.lean.)

Each qubit is a kinematic site for spurious actualization. The cumulative probability of spurious actualization for N qubits:

$$P_{\text{spurious}}(N) = 1 - (1 - p_{\text{th}})^N \approx N \cdot p_{\text{th}} \tag{8}$$

Setting $P_{\text{spurious}} = \eta$ (quality factor) gives:

$$N_{\max} = \eta \cdot p_{\text{th}} \tag{9}$$

Beyond $N_{\max}$, cumulative spurious actualization exceeds the fault-tolerance budget regardless of single-qubit fidelity. Standard QM predicts exponentially difficult but unbounded scaling; RA predicts a hard ceiling.

This is the most near-term falsifiable prediction with strong discriminating power between RA and standard QM.

6 Complete Predictions Table

Table 4: All RA predictions, sorted by experimental proximity. See Papers I–IV for derivations.

ID	Prediction	Status	Experiment/Timeline
P01	BMV null; $\tau_{\text{act}} \approx 1.9 \times 10^4$ s at 1 mK	[DR]	Multiple groups pursuing; near-term
P02	$H_0 \propto \rho_b^{-1/2}$(LOS); 75.9 km/s/Mpc	[CV]	DESI; current data
P03	No DM signal below neutrino floor (categorical)	[DR]	LZ/XENONnT; ongoing
P04	$t^* = 0.274/g$ (spin-bath, parameter-free)	[CV]	SC circuits; medium-term
P05	$N_{\max} = \eta p_{\text{th}}$ (QEC hard ceiling)	[AR]	QEC at $\sim 10^3$ qubits; near-term
P06	Life wherever F1+F2; substrate-independent	[AR]	SETI biosignature search; long-term
P07	T^{-4} vs T^{-1} decoherence scaling	[DR]	Levitated nanoparticle experiments
P08	No proton decay (baryon number topological)	[DR]	Any future proton decay experiment
P09	$\theta_{\text{QCD}} = 0$ exactly; no axion	[DR]	Strong CP experiments; dark matter axion searches

7 Open Problems with Proof Paths

Table 5: Open problems with current best paths to resolution.

ID	Problem	Path to closure
O03	Bianchi on curved background	Perturbative $(l_P/R)^2$ expansion: flat-space term (D04) + curvature correction bounded by LLC range (L01+L02). No Tomita-Takesaki needed at leading order.
O06	l_{RA} from first principles	Dominant $k = 2$ term in D4U02 = minimal BDG diamond; area $= l_P^2 \Rightarrow l_{\mathrm{RA}} = l_P$; gives $\xi = \Delta S^*/l_P^2 = C03/l_P^2$.
O10-G	Lean: $G_{\mu\nu}$ derivation	RA-native route: D04 + O10-spin2 directly; bypasses Weinberg S-matrix theorem.
O11	Lorentz invariance from DAG	BDG modular invariants (c_k mod gcd) as discrete Lorentz-invariants; question for Dowker collaboration.
O-I-1	Wyler fractional correction	Derive group volume ratio from BDG dynamics.
O-I-2	CKM matrix	Cabibbo angle conjecture $\lvert V_{us}\rvert = \sin(2/9)$; full quark mixing from $\mathrm{SU}(3)_{\mathrm{gen}}$.
O-II-1	Born rule derivation	Show $a(v\lvert C) \propto \psi(v)$ follows from BDG action.
O-III-3	Dark energy alternative	Quantify intrinsic graph growth rate as $\Lambda = 0$ cosmological expansion mechanism.

8 Integrity Checks

The following integrity checks (IC) document known scope caveats and potential weaknesses. They are recorded here not as defects to be hidden but as precise statements of what the framework currently claims and what it does not.

IC01: L04 uses $\sigma_0 \propto I$ (finite-dimensional truncation). The physical claim requires Poincaré invariance of the Minkowski vacuum, which is a standard QFT result but not the statement proven in Lean.

IC02: D02 (Unruh resolution) covers modular flow stationarity. The explicit detector click story (no actualization in Minkowski vacuum) is now DR via O05r, which uses L05+timescale formula.

IC03: $l_{\mathrm{RA}} = \sqrt{4\Delta S^*}\ell_P$ is derived from area-law self-consistency (D08), not from BDG geometry independently (O06).

IC04: A01 (GR uniqueness) uses $d = 4$ imported from L11 via RATM/RASM. Should be derived directly in the Paper III context.

IC05: A03 (BMV null mechanism) uses "unactualized $\Rightarrow \rho_A = 0$" which requires O07 for the macroscopic COM case. O07 is now DR, so this is substantially closed.

IC06: C01 (α_s result) gives the UV fixed point at $\mu \to \infty$. The IR fixed point $\alpha_s = 1/3$ is also proved [CV]. The relationship between the two is the BDG two-state RG flow.

IC07: L07 (causal invariance) was conditional on amplitude locality in RA_AQFT_Proofs_v10.lean. O01-O02 in RA_AmpLocality.lean are unconditional. L07 is superseded by O01+O02.

9 The Philosophical Synthesis

9.1 What Relational Actualism Claims About Physics

Physics has operated for four centuries with a foundational metaphor: laws on one side, phenomena on the other. The laws are fixed, background structure; the phenomena play out against them. This structure is so deeply embedded that it is rarely stated explicitly. It is simply how physics is done.

Relational Actualism dissolves this distinction. The Local Ledger Condition (L01) is not a constraint that actualization events obey. It *is* what actualization events are, stated as a bookkeeping property. Conservation of energy is not imposed on the causal graph from outside — it is the LLC read at the appropriate scale. General relativity is not a law governing spacetime — it is the effective description of the actualization density field at scales far above the Planck length.

The arrow of time is not explained by the framework. It is what the framework *is*. Every piece of RA mathematics is time-asymmetric because actualization is irreversible by definition.

9.2 The Web, Not the Chain

The dependency structure of RA is not a derivation chain (axioms to theorems in sequence). It is a web. The LLC (L01) connects to gravitational conservation (D04), quantum information (L03), biological complexity (via D11), and black hole thermodynamics (via D10). The BDG threshold ΔS^* appears in quantum measurement, BMV experiments, biological chemistry, the Unruh effect, the Big Bang, and the origin of life — not as an approximate connection but as the same formula evaluated at different scales.

The open problems (Table 5) are not independent mysteries. Each is the same question — "when does a process become irreversible?" — asked at a different scale. The answer is always $\Delta S^* = 0.60069$, because that question has only one answer when irreversibility is the primitive.

9.3 Coherence as Evidence

The internal coherence of RA — the fact that results at different scales connect through the same constants and the same causal graph structure — is itself evidence of something. It does not prove the framework is correct. A beautifully coherent framework can be wrong, and RA makes specific predictions that can falsify it.

But coherence is evidence. When a single commitment (irreversibility is primitive) simultaneously addresses the measurement problem, the cosmological constant, the Hubble tension, the origin of biological complexity, the number of spacetime dimensions, and

the values of the coupling constants — without free parameters and without adjusting the framework to fit — this is not a coincidence. It is either the beginning of a correct understanding of Nature, or it is the most instructive wrong answer in the history of physics.

Either outcome advances physics.

Acknowledgments

This programme was developed over twenty-three years and formalised over approximately twelve months with AI collaboration (Claude, Anthropic; Gemini, Google). The Lean 4 infrastructure builds on the Lean-QuantumInfo library (Meiburg, Perimeter Institute). All papers are open-access at https://zenodo.org/communities/relational-actualism. Lean 4 files, Python scripts, and figure sources are at https://github.com/jsandeman/Relational-Actualism.

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10 Integrity Checks: Known Scope Caveats

The following integrity checks (IC) document precisely what the framework currently claims and what it does not. They are recorded not as defects but as honest statements of scope. Every theoretical programme has limits; the RA programme documents them explicitly.

10.1 Layer 0 Integrity Checks (Lean-Verified)

IC01 (L04 scope). Theorem L04 (Frame Independence) is proved in a finite-dimensional truncation with $\sigma_0 \propto I$. The physical claim requires Poincaré invariance of the Minkowski vacuum, which is a standard result in algebraic QFT (Haag [**?**]). The gap: finite-dimensional $\sigma_0 \propto I$ is not Poincaré-invariant (it is invariant under all unitaries, not just Poincaré ones). The physical significance is limited: the finite-dimensional result is sufficient for the main application (Lorentz covariance of the actualization criterion, D01), and the extension to the full AQFT setting is expected to hold.

IC07 (L07 status). L07 (conditional causal invariance) is superseded by O01+O02 (unconditional, any permutation). The RA AQFT files contain both. For any paper that cites causal invariance, the unconditional version (O01+O02 from RA_AmpLocality.lean) should be used.

10.2 Layer 1 Integrity Checks (Computation-Verified)

IC06 (α_s fixed points). The computation C01 establishes the UV fixed point $\alpha_s(m_Z) = 1/\sqrt{72}$ at $\mu \to \infty$. The IR fixed point $\alpha_s = 1/3$ at $\mu \to 0$ is also computation-verified but uses the two-state model approximation. Papers discussing the strong coupling should note both fixed points.

IC-ΔS^* (Irrationality). $\Delta S^* = 0.60069$ is established as the exact value to 10^{-10} precision. The rational approximation $3/5 = 0.600$ is ruled out at 25.2σ. Any claim that uses ΔS^* numerically should use 0.60069 (or the more precise value 0.548436... for $P_{\text{acc}}(1)$), not 3/5.

10.3 Layer 2 Integrity Checks (Derived)

IC02 (D02 gap). D02 (Unruh resolution via modular flow stationarity) covers the statement that $\Delta S_{\text{per photon}} = 0$ for the Rindler bath. The full story of "no detector clicks" (O05r) requires the additional step of applying the universal timescale formula. Papers presenting D02 should cite O05r for the complete argument.

IC03 (ξ provisional). $\xi = 1/(4\ell_P^2)$ is derived from the area-law self-consistency $\ell_{\text{RA}} = \sqrt{4\Delta S^* \ell_P}$. This is a self-consistency condition, not an independent derivation. Until O06 is resolved, papers should present ξ as "provisionally derived" rather than "derived from first principles."

IC04 ($d = 4$ importation). A01 (GR uniqueness via Lovelock) uses $d = 4$ imported from L11 via RATM/RASM. The direct derivation of $d = 4$ in the GR context (without reference to the SM topology papers) has not been written.

IC05 (Macroscopic COM actualization). A03 (BMV null mechanism: superposed mass sources zero metric) uses "unactualized $\Rightarrow \rho_A = 0$". For a single quantum particle, this is straightforward. For a macroscopic center-of-mass in spatial superposition, the argument requires that $\tau_{\text{act}} \gg \tau_{\text{exp}}$ for the COM — which is established by O07r. Papers presenting A03 should cite O07r as the timescale closure.

11 The Philosophical Synthesis: Laws Are Phenomena

11.1 The Standard Picture and Its Hidden Assumption

Physics operates with an implicit hierarchy: laws govern phenomena. The laws of nature are the fixed, background structure. The phenomena — particle collisions, chemical reactions, biological processes — play out against this background. This hierarchy is so deeply embedded that it is rarely stated: it is simply how physics is done.

The laws are taken to have a different ontological status from the phenomena. They are not observed; they are inferred. They are not events; they are regularities. A law says something like "whenever event A occurs, event B will follow" — but the law itself is not an event. It is a relation between events, a description of the structure of reality, not a part of reality itself.

This hierarchy has a name in philosophy of science: it is the nomological account of laws (from the Greek *nomos*, law). Nature has laws; the laws govern nature; the laws are not themselves part of nature.

11.2 How RA Dissolves the Hierarchy

Relational Actualism makes the nomological account mathematically untenable.

The Local Ledger Condition (Theorem L01) is proved in Lean 4. It states: at every vertex of the causal DAG, $\Sigma_{\text{out}}(v) = \Sigma_{\text{in}}(v)$. Is this a law or a phenomenon?

It is not a law in the nomological sense: it does not say "whenever an event occurs, this balance holds." It is a property of every event: it is what every actualization event *is*, stated as a bookkeeping property. An actualization event with $\Sigma_{\text{out}} \neq \Sigma_{\text{in}}$ is not a violation of a law — it is a self-contradiction. It would not be an event in the RA sense at all.

The LLC is not a constraint imposed on events from outside. It is part of the definition of what an event is.

The same is true throughout RA:

- Conservation of energy is the LLC at the appropriate scale — not a law imposed on the causal graph but a property of every vertex in it.
- The arrow of time is not explained by RA — it is what RA *is*. Actualization is irreversible by definition. There is no RA formalism that runs backwards.
- The metric is not evolved by the Einstein equations *as a law*; it is what the actualization density field *looks like* at macroscopic scales. The Einstein equations are the effective description of this appearance.

11.3 The Web Is the Theory

The dependency graph of RA is not a derivation chain (axioms at top, theorems at bottom, flowing in one direction). It is a web. Every node has multiple parents and multiple children. The LLC (L01) connects to gravitational conservation (D04), quantum information (L03), biological complexity (D11), and black hole thermodynamics (D10).

The threshold ΔS^* appears in quantum measurement, BMV experiments, biological chemistry, the Unruh effect, the Big Bang, and the origin of life.

Any attempt to read the web as a chain — "first we prove the LLC, then the BDG action, then the field equations, then the predictions" — misses the structure. The chains exist (the dependency graph is acyclic, so there are chains), but they are not the theory. The theory is the web: the set of all the connections, the way the same fact appears in fourteen different physical contexts, the way ΔS^* is simultaneously the measurement threshold and the origin-of-life threshold and the universe-seed threshold.

The mutual implication structure [**?**] is the correct framing. The laws and the phenomena are not in a hierarchy. They are in mutual implication: the laws are the phenomena viewed at the level of pattern, and the phenomena are the laws instantiated at specific events. Neither is more fundamental than the other.

11.4 What It Would Mean for RA to Be Wrong

A beautiful framework can be wrong. RA makes specific predictions that can falsify it:

- If the BMV experiment detects gravity-mediated entanglement, the core claim that superposed mass sources zero metric is false.
- If DESI finds no correlation between H_0 and baryon density along the line of sight, the Hubble tension mechanism is wrong.
- If LZ or XENONnT detects a WIMP, the categorical exclusion of WIMPs fails.
- If a fourth generation of quarks or leptons is discovered, the BDG closure theorem is wrong.
- If proton decay is observed, exact baryon conservation fails.

Each of these is independently falsifiable. The framework does not protect itself from falsification by post-hoc adjustment. If any prediction fails, the failure is a precise statement of what aspect of the framework is wrong, and it points directly to the physics that needs to change.

11.5 The Significance of Internal Coherence

The absence of free parameters is not merely an aesthetic virtue. It makes the framework's coherence epistemically significant.

If a framework has N free parameters, any agreement between predictions and observations could be attributed to parameter tuning. The framework would be consistent with a wide range of observations, and its coherence would be purchased by fitting.

RA has zero free parameters for any of the predictions in Table 4. Every number — $\alpha_{\mathrm{EM}}^{-1} = 137$, $\alpha_s = 0.11785$, $H_0^{\mathrm{Eridanus}} = 75.9$, $f_0 = 5.42$, $t^* = 0.274/g$ — is a consequence of the same five integers that were fixed by the geometry of the d'Alembertian. The coherence is not purchased by fitting. It is either genuine — the framework correctly captures something about the structure of nature — or it is an extraordinary accident.

The probability that fourteen independent predictions all agree with experiment by accident, in a framework with zero free parameters, is essentially zero. The coherence is evidence.